Investigation of Agricultural Uses of Nuclear Waste Heat

-Final Project Report-

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TABLE OF CONTENTS

PAGE

LIST OF TABLES III
LIST OF FIGURES IV
SUMMARY V
CHAPTER 1. INTRODUCTION 1
CHAPTER 2. REVIEW OF PAST WASTE HEAT PROGRAMS AND RESEARCH 4
2.1. Agricultural applications 7
2.1.1. Agricultural Products 8
2.1.2. Non-food Agricultural Applications 9
2.2. Waste heat sources 11
2.2.1. Fossil Fuel Plants 11
2.2.2. Nuclear Power Plants 12
2.2.3. Other Industrial Heat Sources 17
CHAPTER 3. ASSESSMENT OF AGRICULTURAL OPPORTUNITIES FOR EXELON NUCLEAR POWER PLANTS 19
3.1. Waste heat profile of Exelon NPPs 19
3.1.1. Braidwood Generating Station 20
3.1.2. Byron Nuclear Generating Station 21
3.1.3. Dresden Nuclear Generating Station 21
3.1.4. Calvert Cliffs Nuclear Power Plant 22
3.2. Potential agricultural applications for specific Exelon NPPs 22
3.2.1. Greenhouse Horticulture 23
3.2.2. Open-Field Farming 24
3.2.3. Aquaculture Systems 24
3.2.4. Liquefied Natural Gas (LNG) Vaporization 25
CHAPTER 4. ECONOMIC REVIEW OF HEATING APPLICATIONS 27
4.1. Agronomic factors 27
4.2. Economic considerations of waste heat requirements 32
4.2.1. Analysis for different crops 37
4.2.2. Horticulture Scale Analysis 38
CHAPTER 5. REGULATORY CONSIDERATIONS FOR WASTE HEAT USES 43
5.1. Overview of regulatory factors 43
5.2. Considerations for potential agricultural applications 45
CHAPTER 6. RECOMMENDATIONS AND CONCLUDING REMARKS 46
LIST OF REFERENCES 48

LIST OF TABLES

Table Page

Table 1 Benefits and potential challenges of various agricultural applications.....10
Table 2 Summary of some alternative industrial waste heat sources and applications.....18
Table 3 Waste heat information from four Exelon NPPs.....20
Table 4. Location considerations for agriculture near Exelon NPPs 28
Table 5 Average returns on selected crops 29
Table 6 Projected returns from large greenhouses by crop type.....30
Table 7 Heating fuel comparison (EMC, 2013; U.S. Energy Information Administration, 2018). 35
Table 8 Liquid effluent radiation release.....44

LIST OF FIGURES

Figure Page

Figure 1 Histogram of waste heat publications. *Publication year represents literature published within the indicated decade, based on the literature collected for the waste heat from digital and non-digital sources (n=67). 5
Figure 2 Dampierre NPP, Dampierre-en-Burly, France via Google Earth.....6
Figure 3 Timeline representing nuclear and waste heat milestones.....7
Figure 4 Common greenhouse heating application offered by BioTherm Solutions.....24
Figure 5 SCV and ORV LNG vaporization systems (Eisentrout, 2006).....26
Figure 6 Average profit per square foot of production for (a) conventional and (b) non-conventional greenhouse crops.....31
Figure 7 Greenhouse model unit (earthcaregreenhouse.com, 2015).....32
Figure 8 Average temperature at Baltimore Washington International Airport, MD (National Center for environmental Information, 2018).....33
Figure 9 Heat load per greenhouse model unit (2880 ft^2) throughout the year. 34
Figure 10 Annual cost for heating for different fuels per 1000 ft^2 of greenhouse area. Electricity 1 considers an air-source heat pump as the heat appliance while the rest of the cases consider boilers. 36
Figure 11 Cost per square foot for a year for different crops and locations of the units..38
Figure 12 Air temperature, condenser outlet temperature, and hours when conditions to use NPP waste heat to warm up greenhouse are met. (National Ocean Service, 2018) 39
Figure 13 Zoom into Figure 12 graph: March 31st, 2017 – June 7th, 2017.....40
Figure 14 Zoom into Figure 12 graph: September 7th, 2017 – December 12th, 2017 40

SUMMARY

Nuclear power plants (NPPs) produce significant quantities of waste heat due to the low thermal efficiency of the Rankine cycle. This waste heat carries roughly two-thirds of the total power generated at a nuclear power plant. It is responsibly released to the environment as a warm water flow, carrying with it a massive amount of low-grade thermal energy. As of now, this energy is lost from all Exelon NPPs. This waste heat is essentially a free source of energy and utilization projects offer the potential to significantly increase the thermal efficiency and profitability of centralized power generating stations, in particular NPPs.

Agricultural projects, particularly climate-controlled greenhouses, may show the most dramatic benefit from waste-heat utilization. Due to the relatively low temperature requirements of greenhouses, heating costs may be eliminated or cut substantially by the implementation of waste heat. The technical readiness of coupling central power generation station's waste heat discharge to climate-controlled greenhouses is high and has been demonstrated. Implementing this combined heat and power approach to Exelon's fleet of NPPs could offer significant advantages over fossil fueled power plants due to their high capacity factor, and strictly controlled and monitored water discharge.

Exelon NPPs at Braidwood, Byron, Dresden, and Calvert Cliffs hold potential to supply significant agricultural waste heat. Few regulations exist specific to the use of waste heat from NPPs, but the discharge of waste heat from NPPs is strictly regulated, mitigating potential for regulatory hurdles to its use in other industries. Waste heat from Exelon's NPPs is an untapped and potentially lucrative commodity. In the Midwest, the location of Exelon's NPPs offers interesting opportunities for waste heat use in agriculture. The cold Midwest climate is normally a limiting factor in crop diversity and heated greenhouses are an option for overcoming this issue, but the heating costs are a barrier themselves. With cooperation between growers and the Exelon Corporation, waste heat may be

commoditized while also facilitating more efficient and diverse growing operations. Waste heat use is not limited to farming operations. There are a host of industries that can benefit from waste heat use that may be local to other power plants. The Calvert Cliffs Nuclear Power Plant is located ideally to facilitate the farming of oysters with waste heat or even provide heat for liquefied natural gas vaporization.

This report details a preliminary analysis to evaluate the potential for utilizing waste heat from NPPs. From the exhaustive review of past waste heat efforts, agricultural options, and primary economic and radiological factors, the technical readiness is shown to be high and the concept is sound. It is recommended that a detailed analysis should follow that includes: coupling candidate NPPs to specific greenhouse architectures, exploration of high-value crop options in prototypic and bounding event environments, and a detailed economic assessment.

CHAPTER 1. INTRODUCTION

Nuclear power plants (NPPs) produce large amounts of energy, a portion of which is harnessed to generate electricity through a steam turbine cycle. In current designs, roughly two-thirds of the energy produced is dispelled as waste heat in the form of warm water (generally <120 ºF) from the condenser. Power-generating stations currently dispose of this heat to the environment in a controlled, responsible manner. However, there exists the possibility to use the waste heat to drive low-temperature processes of economic value. Increasing the energy utilization of the power plant can increase profits and net efficiency, as well as expand the local economy.

In some cases, waste heat has been used by the utility as an abundant source of essentially cost-free energy. There is also an opportunity to sell waste heat to lower the heating costs of proximal industries. Low-grade heat is generally insufficient to drive industrial or chemical processes, but its availability at a higher temperature than conventional water sources reduces heating costs. In contrast to industrial processes, agricultural operations commonly require temperatures within the range of waste heat. The thermal energy needed in many agricultural industries can be of low quality, and some of the energy from CHP (combined heat and power; often known as co-generation) plants is already being used for agricultural applications internationally. Use of waste heat from NPPs for agricultural processes, such as supplying the heat needs of controlled agriculture (e.g., greenhouses), are unique in the reliability of waste heat production and stand to appreciably mitigate or potentially eliminate heating costs.

Exelon operates a fleet of NPPs in areas that experience long, cold winters. This places Exelon in an advantageous position to not only turn waste heat into a commodity but to become an integral part of the agricultural sector in regions co-located with NPPs. Waste heat utilization offers an opportunity to profit from this largely untapped, abundant

resource while simultaneously co-benefiting and integrating the nuclear and agricultural industries by reducing heating requirements from carbon producing sources. In the case of Exelon NPPs, potentially substantial savings in heating costs can enable climate controlled production (e.g., greenhouse) of high-value perishable crops that are consumed year-round despite the inability of many such crops to grow well or at all in temperate climates (e.g., central and northern US). In addition to economic advantages, the abundance of waste heat means that it stands to fortify growing public interest in and demand for increased food system sustainability. By lowering the operational costs of regional food systems year-round in temperate climates and by avoiding the carbon footprint embodied in transport and heating costs of high-value produce grown with conventional heating sources, agricultural products yielded with the use of waste heat from NPPs offer a unique advantage to meeting such goals. Decreased or 'avoided' carbon footprints afforded by replacing conventional heat sources with waste heat, as well as lowered 'food miles' from regional production (e.g., greenhouse tomato production in winter months), are likely to have favorable public and consumer perception.

In this report, previous waste heat projects and studies have been analyzed to learn how this field has developed over the past decades and to apply that information to Exelon's NPPs. Agricultural applications, especially heated greenhouses, appear to have the most potential for benefit from waste heat and are a focus of the report.

Based on the lessons learned from previous studies, characteristics of Exelon NPPs are summarized and their suitability for waste heat utilization is discussed. Again, this will focus on agricultural applications in greater detail. There are other potential applications for waste heat, however, including liquefied natural gas vaporization, district heating, and simply providing pre-heated water to high temperature processes.

There are a wide range of factors influencing the economics of waste heat utilization, so a simple model is developed for discussion with room for refinement in a future project phase. This report includes a discussion of basic economic factors of using waste heat in a heated greenhouse system based on projected crop yield and expected revenue. The analysis is then extended to different scales of greenhouse production – the most promising agricultural application of waste from both technical and economic

standpoints – to provide a rough estimate of revenue potential to the greenhouse operator. The waste heat supply is also evaluated for discussion of greenhouse size and environmental factors including the temperature requirements of the greenhouse for different crops, the outdoor temperature throughout the year, and the estimated waste heat temperature throughout the year.

The current state of the regulations affecting waste heat are also reviewed. Waste heat utilization from NPPs is not a common practice and has received very little attention in the United States, leading to a lack of regulation. Regulatory analysis is conducted based on existing regulations concerning ground water and water drinkability. While there are few rules regarding waste heat use, different regulatory agencies, such as the NRC and EPA, reserve the right to introduce rules as they become relevant.

The findings in this report offer an encouraging possibility for greatly increasing the use of thermal energy of nuclear power. There have been plenty of successful examples of waste heat utilization in agriculture around the world. This energy is aptly named "waste heat" as it is currently dumped into large bodies of water or dissipated to the atmosphere, but the possibility exists to not only use this energy but to profit from it as well. Admittedly, waste heat energy is not as easily used or sold as electricity, but at the correct site with sufficient demand this heat can not only be sold but broadcast a positive environmental image for all parties involved. Based on the findings of this report, Exelon is well-positioned to not only be a provider of zero-carbon electricity but also reduce the carbon footprint of perishable produce and other products produced from heated greenhouses by selling a constant, reliable source of heat that would otherwise be wasted.

CHAPTER 2. REVIEW OF PAST WASTE HEAT PROGRAMS AND RESEARCH

The use of heat generated by other means, such as decomposition of organic matter, is a 2,000-year old concept and practice (Smith et al., 2016) but its contribution to agriculture has been generally limited by the marginal amounts of waste heat produced. In the past two centuries, industrial processes such as electricity generation have made it possible for waste heat to contribute significantly to agricultural production (Woolley et al., 2018). Over a century ago, Turnbull (1916) proposed what is likely the first published conceptualization of utilizing low-grade industrial waste heat for agricultural applications: "Would it not be possible to make use of this low-grade heat for agricultural purposes, so supplementing our all too scanty summers?". This concept would reemerge in 1970 when the Environmental Protection Agency (EPA) enacted waste heat disposal regulations for steam-electric generation plants (Boersma and Rykbost, 1973; Meekhof et al., 1977), including fossil fuel and nuclear power plants.

The majority of waste heat research efforts occurred in the 1970s and in the U.S. were authored largely by Oak Ridge National Laboratory (ORNL) and Tennessee Valley Authority (TVA), as well as by researchers in Japan. The ORNL specifically created a program on the beneficial uses of waste heat in 1969 in response to the growing concern of thermal pollution (Olszewski, 1978) and often collaborated with the TVA. The EPA and the National Aeronautics and Space Administration (NASA) often sponsored conferences on waste heat utilization. Similar efforts occurred in this period outside of the United States. For example, in the mid-1960s, Japan began conducting experiments utilizing thermal effluent for aquaculture (Yang, 1970).

In addition to a handful of federal research agencies driving waste heat utilization research, two additional themes emerge from the literature. First, integrated agricultural systems that entail two or more conventionally non-proximal systems are commonly

proposed (Beall and Samuels, 1971; Price and Peart, 1973). Proposed systems often consisted of an aquaculture or animal rearing system linked to a climate-controlled greenhouse, where any remaining waste heat would be fully utilized (Boersma, 1970; Togawa et al., 2014). Second, the pioneer literature on agricultural re-use of nuclear waste heat emphasis a similar type of plant-based agriculture – horticulture – and specific plant crop species. Whether proposed or grown in a pilot experiment, traditional horticultural crops (e.g. tomatoes, cucumbers and leafy greens) dominate the majority of greenhouse production systems followed by floriculture (Helgeson et al., 1986; von Elsner, 1984). Relevant publications on nuclear waste heat re-use was found to be largely (64%) white papers, reports, and a combination of digital and non-digital conference proceedings. Limited access to older journal publications not digitalized was a common occurrence. Available online volumes often did not extend back to the 1970s, necessitating a search for non-digital copies of publications on waste heat utilization via university libraries.

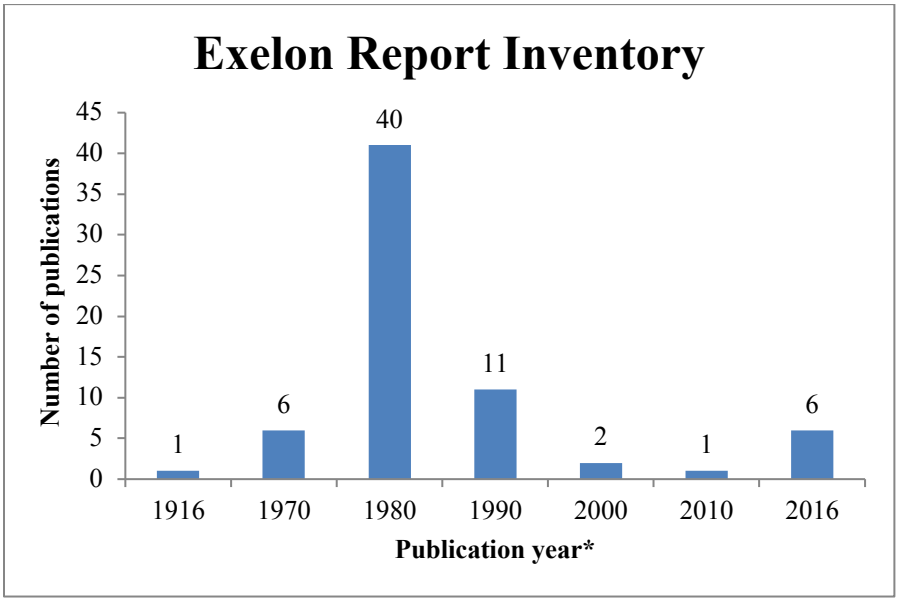

Figure 1 Histogram of waste heat publications. *Publication year represents literature published within the indicated decade, based on the literature collected for the waste heat from digital and non-digital sources (n=67).

A search for visual evidence of agricultural systems paired with NPPs was conducted in Nov. – Dec. 2018 via Google Earth. The United States, France and Japan were selected as areas of interest because these nations account for much of waste heat utilization research identified in the literature review. Glass-paneled greenhouses were the structures of interest and known greenhouse structures were used as basis for visual evaluation of satellite images of NPPs. There was no visual evidence of protected greenhouse structures next to NPPs in the United States and Japan. However, greenhouse systems were identified approximately 350 m from coal-fired plants such as Montour in Washingtonville, PA that was used in a pilot study conducted by Manning (1984). Open-field agriculture and greenhouse systems were found within 50-100 m of the French NPPs EDF CNPE Chinon and Dampierre. Greenhouse systems included glass-paneled structures and polyhouses, which serve a similar purpose as greenhouses (i.e., high-value crop production in cooler months) but may not necessarily be temperature-controlled. It is difficult to determine if the identified agricultural structures employ waste heat from co-located NPPs because the publications require translation from French. This suggests additional knowledge on waste heat re-use for agricultural applications may exist in non-English language literature, in particular Japanese and French.

Figure 2 Dampierre NPP, Dampierre-en-Burly, France via Google Earth

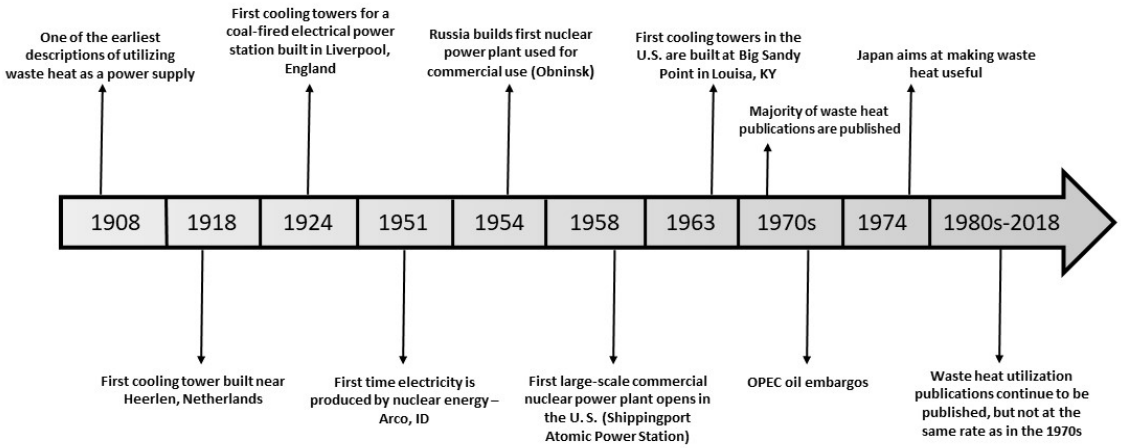

Figure 3 Timeline representing nuclear and waste heat milestones.

2.1. Agricultural applications

Agricultural applications proposed in the identified waste heat literature include: greenhouse systems (Ashley, 1978; Manning, 1984; Ryther, 1977), aquaculture (Coutant, 1970; Pickering, 1970), open-field agriculture with in-ground or irrigation heating (Boersma, 1970; Rykbost, 1975), livestock shelters (Beall, 1971; Miller, 1971), algae production (Boersma, 1970; Bond, 1977) and food processing facilities (Lundberg, 1979). Of these, the dominant system was temperature-controlled greenhouse (n=47 or 70%) followed by aquaculture systems (n=22 or 33%). However, a minority of studies (n=13) entail experimental work with greenhouses to test waste heat re-use at pilot or commercial scales (Burns, 1976; Manning, 1984). Successful preliminary experiments for aquaculture systems are scarcer still (Guerra, 1976; Suffern, 1979), which is likely due to the greater complexity of aquaculture systems relative to greenhouse agriculture (Godfriaux, 1979; Suffern, 1979). Thus, there is a need for experimental work to trial pilot greenhouse and other agricultural operations to determine feasibility of waste heat re-use.

Publications on waste heat utilization for agriculture are not necessarily specific to NPPs. In fact, the majority of identified proposed or pilot studies on waste heat re-use in agriculture evaluate fossil fuel power plants as the heat source. Experimental studies were largely limited to fossil power plants, whereas the majority of nuclear waste heat studies

were theoretical or modeled and lacked detail on agricultural applications (e.g. economics) and do not consider the use of alternative high-value agricultural products. There are fewer pilot studies or scaled-up operations than the theoretical or modeled experiments, which primarily focused on the economic and technical feasibility of using waste heat systems for agricultural applications. Details on the engineering and design of varying agricultural systems are more prevalent than data on crop yield and production parameters. The relative absence of successful nuclear waste heat studies, management and crop data, and consideration for alternative high-value products limits interpretation of past feasibility studies and raises the need for fundamental research to experimentally determine agronomic requirements for and challenges to nuclear waste heat utilization in agricultural systems.

Agricultural industries, particularly the greenhouse sector, currently rely largely on fossil fuels to meet heating demands. Rising fuel costs make it difficult for producers to increase the production capacity of their climate-controlled systems (Manning, 1984). The heating demand of a greenhouse system largely depends on crop type and climate. Waste heat generated from steam-electric power plants offers agricultural industries an opportunity to decrease heating costs and enable production of crops with high heat demands, both of which stand to translate to economic benefits. The utilization of waste heat also enables non-traditional and high-value agricultural products as well as non-food agricultural applications.

2.1.1. Agricultural Products

A variety of horticultural products can be produced in waste heat systems, particularly greenhouses where multiple harvest cycles are achievable. Horticultural crops generally hold a higher return on investment per unit area compared to agronomic crops such as grains (Acquaah, 2009). The literature on waste heat use in agriculture mostly consists of traditional horticultural crops, but also non-conventional greenhouse crops such as spices and herbs (Yu, 2016), floriculture (Johnson, 1982; Manning, 1983; von Elsner, 1984), berries (Boersma, 1973) and vegetables (CASE, 2009; van der Horst, 1972; Piles,

1978). Candidate horticultural products not found within literature include spices and herbs, (sub)tropical crops, medical cannabis and mushroom production.

Climate-controlled greenhouses are likely the most realizable and profitable agricultural system for Exelon's fleet of NPPs. Aquaculture systems will also be considered for Calvert Cliffs due to its proximity to large bodies of water and the prevalence of oyster production in the area. Candidate crops selected for greenhouse production include spices and herbs, mushrooms, (sub)tropical crops and medical cannabis. The crops selected are ideal due to their high heating requirements and have comparable or more rapid timescales to conventional greenhouse crops, except for certain tropical/sub-tropical crops. Additional work is required to understand the agronomic and economic feasibility of non-traditional food crops. Consideration of high-value horticultural products for greenhouse operations is not only essential for maximizing profits, but also to meet demands for local markets that are otherwise met largely imported into the United States (e.g. spices).

2.1.2. Non-food Agricultural Applications

Waste heat systems can be utilized for agricultural products aside from food crops. Potential non-food agricultural applications include crop processing (Togawa, 2014), heating livestock shelters (Beall, 1971; Bond, 1977; Miller, 1971), wastewater treatment, floriculture, and algal production for biofuel (CASE, 2009). The feasibility of non-food production alternatives has little to no experimental or evaluation but is considered to have high profitability potential (CASE, 2009; Yarosh, 1972).

Table 1 Benefits and potential challenges of various agricultural applications

Application	Advantages	Challenges
Greenhouse heating	• Controlled environment • Multiple methods of heat transfer: underground pipelines, finned tubes/pipes, and evaporative pads • Wide range of commodities/some with high demands • Integrated agricultural systems • Climacteric crops can have a longer maturation period if sold locally • High security status due to location near NPP • Humidity is less of an issue at large-scale • Lengthens growing season • Can aid USDA food programs • Can serve as a research facility for local universities and industry	• NPP outages for maintenance/inspection (1-2 months) • Seasonal temp variations of waste heat • Controlling high humidity • Will likely need a back-up system (fossil-fuel powered) • Potential contamination • Limited usage during summer months • Establishing a representative floor temperature can be difficult • Heating and lighting requirements of certain crops may limit year-round production • Irrigation water source • Consumer perception of risk
Aquaculture	• A solution to overfishing • Demand is high • Yield can significantly improve by raising water temps slightly • Wide range of fish/marine species	• Pollution (waste, excess feed) • Disease outbreaks • Feed can be expensive • Coastal systems are more cost effective
Animal shelter heating	• Swine and poultry production in the Midwest • Can possibly capture CO_2 and sell it • Can increase growth rates • Integrated system approach	• Location is influenced by labor costs and climate • Disease control and waste disposal
Soil heating (open-field agriculture)	• Promotes germination and early crop growth stages • Lengthens growing season	• Small-scale operations only • Placement of waste heat tube network may interfere with field management
Biofuel production	• Sustainable fuel source • Commercial algae production systems mainly serve the dietary supplement market (closed systems) • Certain species of algae can grow low water temperatures (15 °C), but are half as productive than other species	• Sensitivity of species to other species • Systems require lots of space • FDA guidelines • Supplemental CO_2 is needed • Light is the limiting factor
Post-production	• Perception of risk is lower • Potential for industrial parks • Minimize transportation costs	• Seasonal temperature variations of waste heat and ambient • Potential contamination
Spirulina production	• Can utilize marginal land • Dietary supplement market	• Most production occurs in open-air raceways • Infrastructure is expensive • Perception of risk
Waste water treatment	• Biophotoreactors can be placed in greenhouses if necessary • Multiple uses (e.g., waste water treatment, fertilizer, specialty chemicals, bioplastics)	• Location specific • Competition between species may occur which may lead to loss of algae

2.2. Waste heat sources

Large-scale waste heat is available from several sources. Fossil fuel and NPPs are reliable generators of waste heat, but waste heat can also be taken from non-energy generation industries like paper mills and garbage disposal (e.g., burning) plants. As discussed in detail below, NPPs provide a constant, reliable source of waste heat useable in agricultural processes while fossil fuels provide similar though less consistent and abundant waste heat. Fossil fuels may also be a source of CO_2 enrichment for greenhouse growing operations, though this may be matched by other sources used in conjunction with a boiler. Industrial processes vary widely in the quality and quantity of heat but feature some common themes that make them more difficult to rely on as compared to power generating stations. While waste heat is widely available, what applications may be considered feasible will greatly depend on the waste heat source. For example, different examples of successful agricultural systems coupled with power plants of different kinds are discussed below, but waste heat from an oil refinery is more logically recycled through the refinery itself to reduce heating costs. There are clear benefits and drawbacks to each waste heat source explored in this section.

2.2.1. Fossil Fuel Plants

Fossil-fuel power plants supply most of the electricity generated in the United States. The abundance and efficiency of these power plants make them a worthwhile source of waste heat for consideration. Waste heat utilization using energy from fossil fuels has been extensively studied for decades, showing that the waste heat can be utilized successfully, but there are hurdles in its deployment and sustainability.

Fossil fuel plants provide many benefits for waste heat utilization. Traditionally, energy provided from fossil fuels is relatively inexpensive compared to the energy generated by other sources. This has been the primary lifeline of fossil fuels in an economy that is increasingly environmentally-conscious and wary of the pollutants from fossil-fuel power plants. However, waste heat utilization for the use of greenhouses may be able to help foster positive public perception of the environmental impacts of fossil-fuel plants

because generated CO_2 can be used to enrich the air within the greenhouse to stimulate the growth of certain crops (Hare, 1984; de Zwart, 1997). This can improve greenhouse productivity while utilizing an additional waste product. CO_2 enrichment is normally achieved through the burning of nonsulfurous fossil fuels in boilers; fossil-fuel plants may need additional emissions controls to avoid also enriching the greenhouse with undesirable pollutants (Hare, 1984).

Waste heat utilization from fossil fuel power plants has been proven to be technologically viable for use with greenhouses. Waste heat is cool enough that typical heated greenhouse designs may be insufficient, but with use of redundant heating and insulating strategies the waste heat warms the greenhouse sufficiently in all seasons (D.R. Mears, 1982). Waste heat from fossil fuel plants tends to be at a similar temperature as that from NPPs (CASE, 2009). Fossil fuel plant waste heat is not as easy to harness for agricultural production because the waste heat temperature and availability are dependent on the power output of the power plant. An advantage to fossil-fuel plants' electricity generation is the capability for load following, but this impedes their usefulness as waste heat sources. Load following causes heat to not be consistently supplied or at a constant temperature, especially in areas with significant solar or wind power. The challenges posed by load following can be mitigated by timing. During colder seasons, such plants tend to have more demand, increasing the available heat load for the greenhouses (Yu, 2016). However, in deregulated energy markets, fossil fuel plants with waste heat supply to greenhouses have increasingly felt burdened by a contracted supply of waste heat which is not always in phase with high electricity prices. Unlike load-following fossil plants or intermittent power sources, nuclear power provides a constant and reliable source of heat.

2.2.2. Nuclear Power Plants

NPPs contribute the largest share of carbon-free electricity in the United States. However, the industry faces heavily regulated operation and struggles to remain economically competitive in de-regulated electricity markets with fossil-fuel power plants, exacerbated by the rapid decline in natural gas prices. Many NPPs have uncertain futures

and increasingly rely on local and state subsidies. Waste heat utilization can fortify the economic entrenchment of NPPs in local economies by providing significant advantages to nearby agriculture and industry.

NPPs provide an immense amount of waste heat that may be considered inexhaustible for all practical purposes: the heat discharged from a 1000 MW_e reactor can supply waste heat to about 10,000 acres of greenhouses (CASE, 2009, Davis, 2010; Miller, 1970; Madewell, 1975). This assumes that all the thermal energy of the waste heat discharge can be captured, yet the actual utilization is highly dependent on design of the greenhouses and environmental conditions. The abundance and constancy of nuclear heat from a single power plant make it well worth consideration for waste heat utilization projects.

Waste heat from NPPs has not been well utilized in the United States, though it has been studied. In Switzerland and France there are successful waste heat utilization projects in the nuclear power industry. In Beznau, Switzerland, the waste heat from an NPP has been successfully used in a combined heat and power system (CASE, 2009). In Grenoble, St. Laurent, and Cadarache, France, separate experiments to use NPP waste heat in greenhouses were successful (P. Balligand, 1978). The combined heat and power system in Beznau is a successful technical demonstration of the ability of nuclear power to provide domestic heat. However, the economic returns were low due to the plant's location in a low-population area with not much industry (CASE, 2009). The project in Beznau shows the usefulness of such a design but also provides a valuable lesson regarding the implementation of such systems.

While waste heat from NPPs is often available at fairly low temperatures, ranging from 60 °F in winter to 110 °F in summer (Madewell, 1975; Olszewski, 1979), it is still proven to be useful in agriculture. Greenhouses using this waste heat have been demonstrated or simulated to be capable of successfully growing tomatoes, cucumbers, lettuce, roses, and chrysanthemums (P. Balligand, 1978; J. Rhyther, 1977; B. J. Bond, 1974). The Grenoble greenhouse used an unorthodox soil heating system wherein waste heat was supplied from the Siloe research reactor (Balligand, 1978). The Siloe reactor provided heated water between 61°F and 79°F to a double-walled plastic mulch placed

directly over the soil at low speed and pressure. This concept was successful but illustrated that heating concepts using the low-temperature waste heat of NPPs may require non-conventional greenhouse designs that take advantage of the abundant heat supply rather than focusing on efficient use of the heat. It is important to note that this system supplied some extra heat to the waste heat water via a heat pump. This allowed for improved control of the greenhouse temperature. A similar greenhouse project in St. Laurent yielded successful results as well utilizing the waste heat from the St. Laurent des Eaux nuclear power plant rather than a research reactor (P. Balligand, 1978).

In France, there is a current waste heat greenhouse project at Bugey, a 3600 MWe, 4 unit NPP (IAEA, 2000). The Bugey greenhouse project is over 12 acres and was established in 1984 to grow potted plant species (CEC, 1988). Currently operating greenhouse coupled to a nuclear plant, Bugey is a valuable case study in how such a project may be implemented from start to finish, though it should be noted that it was established in a different economic and regulatory environment. The Bugey greenhouse project began as the vision of eight horticulture growers who wanted to heat greenhouses with a geothermal well, but found that heat source to be insufficient and instead sought to procure waste heat from the Bugey power plant. This project was realized by an agency, SAFER, acquiring the land and servicing it, including arranging for the supply of waste heat water. SAFER would then sell lots to different greenhouse growers. A non-profit organization, AUTEC, performed the technical, commercial, and financial studies that the project needed to expand. The horticulturalists then created a cooperative to coordinate production and commercialization of their products while retaining individual responsibility for production. During the construction of the greenhouses, the growers progressively began raising crops so that at the end of greenhouse installation the first plants were already being sold. The project was hit with financial difficulties at the outset, however, due to overestimating the average turnover per area, leading the cooperative to declare bankruptcy only eight months later. The greenhouses were purchased by the Le Bugey Horticultural Company, majority owned by Geotherma, another eight months later which was able to utilize the greenhouses for ornamental plant production (i.e., floriculture). The greenhouse is less than one mile from the power plant and receives waste heat temperatures from 68°F

to 93°F. Heating is achieved by heating carpets to maintain temperatures above 64°F. Like the other heated greenhouses discussed, this one includes a back-up heating system. It is the opinion of all stakeholders that initial struggles at the Bugey greenhouse were due to the first-of-a-kind nature of the project and the use of a much larger greenhouse than most French horticulturalists had experience operating. The initial financial struggles were not due to the use of waste heat and would have in fact been made worse without the savings from waste heat use (CEC, 1988).

In addition to controlled environments such as greenhouses, waste heat from NPPs has proven potential to bolster open-field farming productivity. In a study of open-field farming with waste heat in Cadarache, France, nearly 50% greater soybean yields were achieve with heated pipes using NPP waste heat compared to unheated soil. While this study showed the technological feasibility of using waste heat in buried pipes, it was found to be prohibitively expensive. Due to the high costs, this should only be considered for high-value crops (P. Balligand, 1978).

Due to the low temperature of NPP waste heat, temperature loss over distance is a major issue of concern in greenhouse applications (Lee, 2015). Greenhouses should be located near the power plant to mitigate heat losses. There is precedent in the United States for reserving land within the exclusion zone for an experimental waste heat agricultural system which significantly helps the issue of transporting low-grade heat over distance (B. J. Bond, 1974). It should be noted that at NPPs that utilize cooling towers, waste heat temperatures may be raised by 20°F to 40°F (Madewell, 1975).

To make the waste heat water easier to use and increase the range at which it may be used, the warm water may be heated to higher temperatures (Miller, 1971; Yu, 2016; P. Balligand, 1978; Boersma, 1974; Rotz, 1979; Lee, 2015). In any waste heat system, heat pumps or boilers are typically in place already to control the temperature and act as a bulwark against loss of waste heat. In greenhouse operations, back-up heating systems are common to protect against crop failure. There are significant benefits to preheating the waste heat which are anticipated to justify the costs and added complexity (Hare, 1984). If the water is heated with a nonsulfurous fossil fuel, the resulting CO_2 can be used to enrich greenhouse atmospheres, leveling the potential advantage of waste heat derived from fossil

fuel plants relative to NPPs (Hare, 1984). Boiler-assisted heating of warm water is shown to be the most efficient means of preheating waste heat (Rotz, 1979). Heating can also be achieved electrically if a zero-emission system is desired at the cost of CO_2 enrichment in the greenhouse.

The United States has a more limited history with utilizing NPP waste heat but has studied the concept in depth through TVA and ORNL with the intention of using waste heat from the Browns Ferry Nuclear Power Plant. Experiments conducted with simulated reactor waste heat in Muscle Shoals, Alabama, used a different waste heat design than the one used in the French experiments. The ORNL greenhouse design utilizes evaporative pads which heat the greenhouse with low-temperature water like the Grenoble design but raises greenhouse humidity. This design shows that simpler heating solutions exist than those used in French experiments, but the heating designs chosen may impact other conditions adversely. It is possible to control humidity with additional equipment such as finned tube heat exchangers (Madewell, 1975; Rotz, 1979).

The Canadian nuclear industry offers lessons on utilizing waste heat generated by NPPs for collocated and largely agricultural industries. Bruce Nuclear Generating Station, while not a supplier of waste heat, has been used to provide steam for years. CANDU reactors at Bruce were designed with undersized turbines to allow for the cogeneration of steam diverted to a heavy water production facility. When heavy water production was no longer necessary at Bruce, steam was diverted to an industrial park. The industrial park is 245 acres and operates based on contracted low-cost energy in the form of steam from Bruce. In the industrial park, steam can be used to drive a number of processes, including space heating, curing, hydrolysis, dehydration, evaporation, concentration, and distillation. There is currently an alfalfa dehydration cubing plant, a plastic film extrusion plant, an 8-acre greenhouse equipped with hydroponics, and a commercial alcohol plant (BEC, 2011). While the Bruce Energy Centre still exists and operates on a similar model, Bruce Nuclear Generating Station is positioned to sell electricity at a high enough cost that it made more sense to replace the turbines in order to increases electricity production and forego supplying steam.

2.2.3. Other Industrial Heat Sources

In some cases, industrial heat sources have been considered for waste heat utilization outside of power generation. Many factories and refineries appear to be attractive waste heat sources if they have an abundance of process heat as this is available at much higher temperatures than the waste heat of nuclear and fossil fuel plants (Lee, 2015). Most industrial processes produce a sizeable amount of waste heat, and waste heat is produced by many non-electricity generating industries.

These non-NPP producers of waste heat have begun to realize the potential of waste heat in various applications that serve as case cases illustrative to NPPs. For example, garbage incineration plants generate waste heat that tends to be about 115°F, a useful temperature for greenhouse heating. Successful floriculture (roses) has been achieved using waste heat from garbage incineration plants in Germany, resulting in significant savings for the grower (Von Elsner, 1984). Another example are oil refineries. Though oil refineries generate a large amount of heat at temperatures commonly well above waste heat temperatures which gets recycled through heat exchangers, but some heat inevitably exits as heated water. The Chevron, Inc. oil refinery in El Segundo, CA invested in waste heat boilers (Chevron, 2008) which increases the refinery heating efficiency while lowering heating costs and addressing environmental concerns of coastal California. In Sweden, as tech companies seek to appear more environmentally conscious, data centers in the Stockholm area have been using waste heat for space-heating. Stockholm's Interxion 1600 m^2 data center achieved an 80% reduction in its energy costs after introducing a novel cooling system using sea water and providing waste heat for offices and residences before returning water to the sea. (Oró, 2015). Waste heat from a pulp and paper plant has been used near Springfield, OR for open-field heating. Waste heat water was available at temperatures ranging from 90°F to 130°F and pumped over 2 miles to hydrants where growers could utilize the waste heat as a spray to warm crops. This did not precisely control temperature like in a greenhouse system but successfully protected the crops from the onset of frost (Miller, 1970).

While use of waste heat from industrial sources for farm use has been explored and deemed technologically viable, there are significant hurdles to implementation. One

anticipated obstacle is that industrial and fossil fuel sources of heat often have more down time than power plants. This is analogous to the load-following challenge of fossil fuel plants but more severe as the heat source may be absent much of the time and does not necessarily change favorably with weather. The Springfield, OR pulp and paper mill worked around this hurdle by supplying waste heat for a process that was only needed at certain times in the year. A year-round heating system may be supplemented with the heat from a factory, but the system would most likely be heated by boilers for long periods of time. The feasibility of utilizing waste heat from industries outside of power production varies on a case-by-case basis depending on the quality, abundance, and consistency of heat.

Table 2 Summary of some alternative industrial waste heat sources and applications

Industry	Temperature	Demonstrated Uses
Garbage Incineration	~115°F [1]	Floriculture [1]
Pulp and Paper	90°F-130°F [2]	Open-field heating via hydrant [2]
Oil Refinement	N/A, varies by refinery	Heating cost reduction (process heat) [3]
Data Centers	122°F-140°F [4]	District heating [4]
[1]Von Elsner, 1984; [2]Miller, 1970; [3]Chevron, 2008; [4]Oró, 2015		

CHAPTER 3. ASSESSMENT OF AGRICULTURAL OPPORTUNITIES FOR EXELON NUCLEAR POWER PLANTS

Waste heat utilization is technologically achievable, but whether it is practical and ultimately profitable depends on conditions local to the heat source. Exelon's extensive fleet of NPPs in the Midwestern United States holds high potential to significantly contribute to and potentially transform greenhouse production heating needs and crop choices, respectively. Agriculture is a dominant industry in the Midwest and lowered heating costs throughout waste heat reuse stands to lower production costs for growers, additional and novel crop options while opening new markets and fostering an environmentally-friendly image for NPPs by the public. In addition to crop agriculture at the coastal Calvert Cliffs NPP, there is also the option for aquaculture and providing heat for animal (e.g., poultry) rearing operations as well as nearby non-agricultural industries. The challenge to achieve this vision lies in choosing a suitable location for a waste heat project, designing such a project to be profitable and attracting growers willing to take advantage of the project.

3.1. Waste heat profile of Exelon NPPs

Four Exelon NPPs are evaluated in detail for their potential for agricultural use of waste heat: Braidwood, Byron, Dresden, and Calvert Cliffs. Each of these NPP produces waste heat that can be used for other purposes. Vital information for assessing the feasibility of agricultural waste heat utilization at these sites includes the minimum remaining lifetime of the reactor, the usual duration of NPP downtime, the inlet and outlet temperatures of the condenser cooling fluid, and the flow rate of cooling water out of the condenser.

Table 3 Waste heat information from four Exelon NPPs

	Braidwood Generating Station	**Byron Generating Station**	**Dresden Generating Station**	**Calvert Cliffs Nuclear Power Plant**
Units	2	2	2	2
Location	Braceville, IL	Byron, IL	Morris, IL	Lusby, MD
Owned Land Acreage	4,457	1,398	953 + 1274 (pond)	1,135
Year License Expires	2046, 2047	2044, 2046	2029, 2031	2034, 2036
Licensed Thermal Power [MWth]	3645	3645	2957	2737
Design Electrical Rate [net MWe]	1120	1120	794	845
Reactor Type	Westinghouse Four-Loop PWR	Westinghouse Four-Loop PWR	General Electric Type 3 BWR	Combustion Engineering PWR
Coolant Source	Kankakee River	Rock River	Kankakee River	Chesapeake Bay
Inlet and Outlet Min/Max Temperature (^{o}F)	Inlet: 33.4/81.3 Outlet: 60/90	Inlet: 32.2/74.1 Outlet: /97	Inlet: 40/85 Outlet: 60/90	Inlet: 34/84 Outlet: /<120
Design Condenser Temperature Rise (^{o}F)	21	24	23	10
Inlet/Outlet Flow Rate x10^{3} (gpm)	730	632	471	1200
Thermal Power Lost (MW)	2250	2600	2163	1760
NRC, 2013(a); NRC, 2013(b); AEC, 1974; AEC, 1973				

3.1.1. Braidwood Generating Station

Braidwood Generating Station is an attractive option for waste heat utilization due to its extensive footprint of 4457 acres (NUREG-1437 Vol. 1, Part 2). With a license through 2046, waste heat projects located at the Braidwood facility could run for decades and become an integral part of the regional economy. Planned outages in the near future will last for 18 and 27 days. Running a redundant heating system in the case of greenhouse horticulture for this length of time should be simple, though it may not even be necessary depending on the season, the design of the greenhouse, and crop choice. If upcoming

outages are taken as common practice, outages are short and not planned for the winter when heat needs are greatest. The main condenser temperature rise is designed as 21°F (NUREG-1437, Vol. 2), resulting in outlet temperatures between 60°F in the winter and 90°F in the summer (Braidwood Station, Units 1 & 2, Final Environmental Statement). These temperatures are low enough that they may require some preheating before use depending on the design of the greenhouse and crop choice. The waste heat discharged from the main condenser at 730,000 gpm (NUREG-1437, Vol. 1, Part 2) is more than can be realistically used in collocated agricultural industries (2250 MW).

3.1.2. Byron Nuclear Generating Station

Byron Generating Station is located on a 1398-acre property (NUREG-1437, Vol. 1, Part 2) which leaves sufficient room for land-intensive waste heat projects like greenhouses. The Byron license is valid until 2044, leaving a considerable amount of time for waste-heat-utilizing industries to grow more important to the region and more reliant on Byron. Byron's outages in the near future are not in the winter and planned to last for 26 and 19 days. The main condenser temperature rise of 28°F (NUREG-1437, Vol. 2) results in an estimated discharge temperature of 56°F to 97°F. The waste heat is discharged from the main condenser at 632,000 gpm (NUREG-1437, Vol. 1, Part 2). Byron discharges roughly 2600 MW of thermal energy as waste heat.

3.1.3. Dresden Nuclear Generating Station

Of the four examined NPPs, Dresden Generating Station is located on the least land area (953 acres). Dresden's license is also due to expire sooner, in 2029. Dresden, does however, dissipate a large quantity of heat with a substantial 23°F temperature rise. Dresden dissipates about 2100 MW of thermal energy as waste heat. From these figures, Dresden does not seem to have any advantages over the other NPPs, but its exceptionally low reported release of radiation discussed later may make it the easiest to use for agriculture from a regulatory perspective.

3.1.4. Calvert Cliffs Nuclear Power Plant

Calvert Cliffs Nuclear Power Plant sits on 1,135 acres in Maryland. (NUREG-1437, Vol. 1, Part 2). It is the only plant considered outside of Illinois, allowing waste heat projects to be considered in a slightly warmer climate. The license at Calvert Cliffs expires in 2034. Calvert Cliffs has an upcoming outage planned to take 22 days. The main condenser is designed for a temperature rise of 10°F (NUREG-1437, Vol. 2), which is substantially less than the other NPPs considered. Calvert Cliffs can discharge waste heat directly into Chesapeake Bay due to the low temperature rise. Discharge temperature could not be found but are likely in the range 44°F to 94°F based on the inlet temperature and designed rise (NRC, 2009). These lower temperatures are achieved in part by a high flow rate, 1,200,000 gpm (NUREG-1437, Vol. 1, Part 2). A waste heat project at Calvert Cliffs would likely require preheating of the condenser discharge or a redesign of the heat disposal of the plant itself. Calvert Cliffs produces the least waste heat of these three NPPs at 1760 MW, though for agriculture purposes still represents a vast reserve of unused energy.

3.2. Potential agricultural applications for specific Exelon NPPs

Waste heat utilization has been proposed as a cost-saving and environmentally-conscious measure for several recipient industries. Even industries requiring high-grade heat can save on total heating costs and reduce emissions by partial use of waste heat. The greatest advantages can be realized by industries that are able to directly use the waste heat. Biological applications such as agriculture often require temperature ranges achievable with waste heat.

Waste heat projects do not necessarily need to be limited to these relatively low temperature processes. For environmental reasons, it is important for waste heat to be discharged at a temperature that will not have a significant impact on the area. If waste heat is intended to be used in other processes, it is conceivable that there would be benefits to redesigning how heat is rejected from the power plant in order to obtain higher temperature waste streams and/or at a lower flow rate. Without redesign of the power generating system, waste heat can be used but is limited to small subset of processes.

3.2.1. Greenhouse Horticulture

Greenhouses have been proven to be operable with waste heat available at temperatures from waste heat streams generated from Exelon NPPs. Without modification to the power plant, some amount of preheating will most likely be required to consistently maintain greenhouse temperatures if traditional greenhouse crops and approaches are implemented. Alternatively, the heat disposal of the power plant may be modified to produce higher temperature waste heat. This would most likely affect design of the condenser and result in a lower flowrate of waste water. A lower flowrate should not adversely affect a greenhouse operation as there is presently a higher flowrate than can be realistically used, but the lower flowrate may raise regulatory concerns by increasing tritium concentrations and potentially result in dumping heat to the environment at temperatures beyond regulatory limits.

Two of the NPPs examined herein (Braidwood and Byron) are located in Illinois in high proximity to cropland. While the waste heat temperature they provide may not satisfy a greenhouse's requirements throughout the year, this can be remedied with additional heating or an unconventionally designed greenhouse, depending on the crops grown. While greenhouses growing traditional crops (e.g., lettuce, tomatoes, cucumbers) have been successfully operated with waste heat from NPPs, it may be desirable to try growing non-conventional high-value crops that would otherwise be imported or can sell for higher profit. This would do more to promote the environmentally-friendly nature of NPP waste heat by allowing growers to tout products that are commonly imported as locally-sourced and having a low carbon footprint.

Common commercial heating applications for greenhouse production include in-ground heating, benchtop and overhead devices. Cooling apparatuses such as evaporative pads also use a heat source to produce cool air for controlling humidity levels within a greenhouse. Food crops would not come into direct contact with the supply of waste heat.

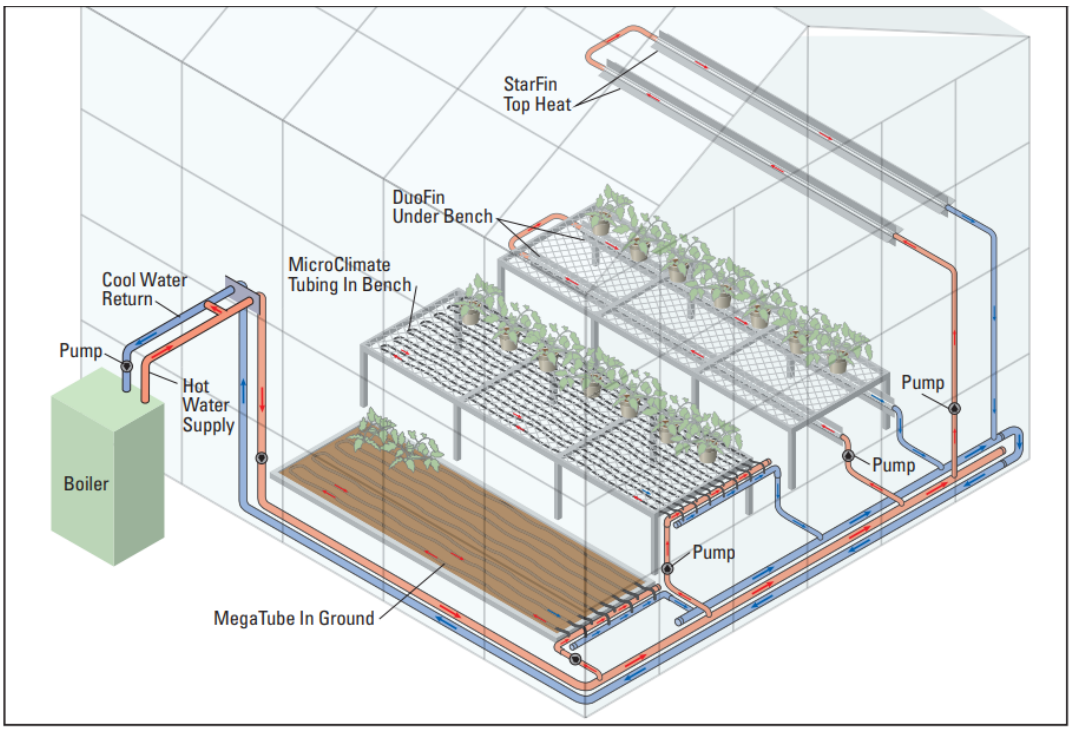

Figure 4 Common greenhouse heating application offered by BioTherm Solutions

3.2.2. Open-Field Farming

Open-field farming is another agricultural application for waste heat in the absence of greenhouses. Open-field heating is achieved through buried, heated pipes and/or warm water-spraying hydrants. Buried pipes achieve constant heating and are more useful for acclimating NPPs that may not typically grow well in the climate (P. Balligand, 1978). Spray from hydrants can be used to warm crops and can protect against frost-induced economic losses but does not offer the same degree of temperature control that greenhouses or even buried pipes can offer (Miller, 1970).

3.2.3. Aquaculture Systems

Aquaculture is an important consideration for the Calvert Cliffs Nuclear Power plant due to its geographical location. Several oyster production systems exist in Maryland and Calvert Cliffs sits in the middle. Coastal systems for aquaculture systems are ideal

because it is more cost effective in terms of infrastructure. Also, shellfish yield can be significantly increased by raising water temperatures slightly (Yarosh, 1972). If an integrated agricultural operation is used, waste heat could first be used in an aquaculture system and then transferred to a climate-controlled greenhouse for subsequent crop production.

3.2.4. Liquefied Natural Gas (LNG) Vaporization

LNG vaporization is an exception to the general trend of industrial processes requiring temperatures beyond what can be provided through waste heat. This option is far more restrictive than others discussed so far in terms of location as the waste heat source must be located near a facility that prepares natural gas. The Calvert Cliffs power plant is located only three miles from Dominion Energy Cove Point, an LNG shipping terminal with LNG vaporization.

Such a waste heat project would require cooperation from Dominion Energy, but such an arrangement could be beneficial to both Dominion and Exelon. Cove point operates 10 submerged combustion vaporizers (SCVs) (MD Dept. of Environment) which vaporizes LNG quickly at high temperatures by burning some of the gas, resulting in some lost product. Open rack vaporizers (ORVs) are the more common system and offer advantages to waste heat. In an ORV system, sea water is run over aluminum-alloy tubes to provide heat from the environment rather than from a furnace. This process is effective and can replace or supplement SCV systems, but has some dependence on sea water temperatures and cools the surrounding water which may impact the environment adversely (Hisada, 2004). The sea water in ORV systems can be replaced by waste heat. This would provide a faster vaporization process than traditional ORV to Dominion Point and neutralize temperature change concerns of Dominion's and Exelon's operations. Dominion Point's current SCV system could then operate to meet fluctuating demand while the new waste-heat supplied ORV system would provide a cheap baseload vaporization capacity. While there are clear advantages to such a system, it would present some engineering challenges requiring cooperation between Exelon and Dominion.

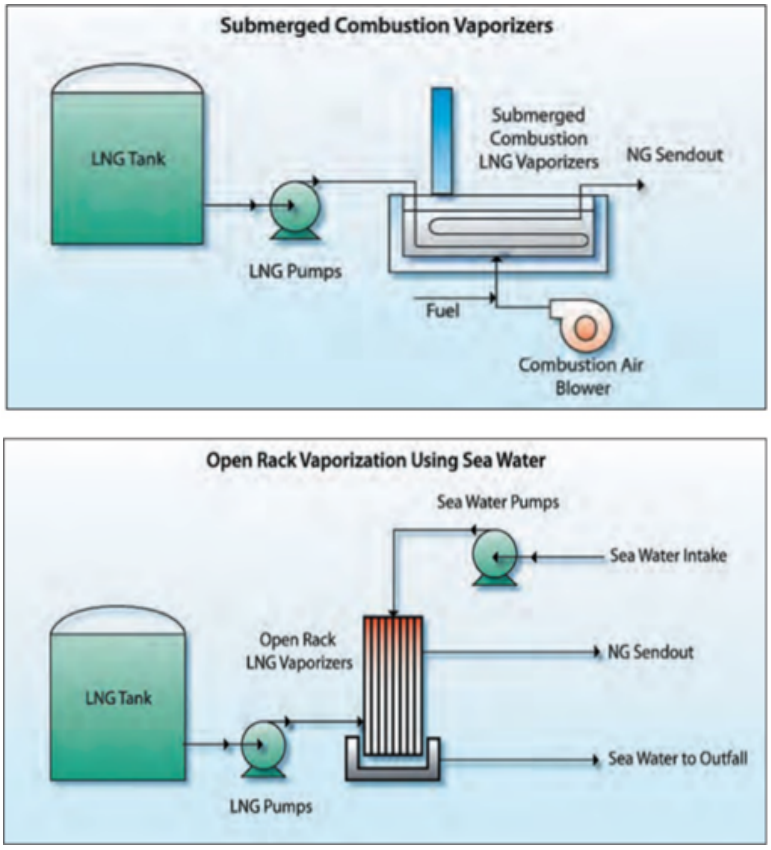

Figure 5 SCV and ORV LNG vaporization systems (Eisentrout, 2006)

CHAPTER 4. ECONOMIC REVIEW OF HEATING APPLICATIONS

The commercial success of a waste heat greenhouse is affected by agronomic factors as well as the energy requirements of the greenhouse. The profitability of greenhouse operations is determined by crop choice, how much of a return those crops can be sold for, and the demand for those crops. For this reason, we consider the growing conditions at proposed greenhouse locations, typical values of common greenhouse crops, and the proximity to urban centers (i.e., markets). Weather conditions around the year are considered to determine what energy needs a greenhouse may have. Heating costs from different fuels are also compared to gauge the economic necessity for waste heat as a source.

4.1. Agronomic factors

Heat is the main agronomic challenge and financial cost to greenhouse production (NGMA, 1998). The feasibility of waste heat utilization for agricultural applications is dependent on a variety of complex environmental and agronomic factors, though general assessments of technical and economic feasibility are possible. The mean annual temperature of a geographic location will influence the heating demands of an agricultural system. An additional factor requiring consideration includes crop type, which will determine planting density, temperature and lighting requirements, harvest potential, management requirements and anticipated use of the crop. Precise economic estimates for the candidate crops are challenging due to limited availability of greenhouse crop production data and knowledge on growing non-traditional horticultural products. Location of crop production systems and market demand ultimately sets the economic foundation of

a cropping system. However, there are general and strong differences in market price values of candidate crops.

Table 4. Location considerations for agriculture near Exelon NPPs

Power Plant	Location	Mean Annual Temperature (2007-2017)	Proximity to urban center (population)	Mean Annual Growing Degree Days (2007-2017)
Braidwood Generating Station	Braceville, IL	49.6 °F	Dekalb – 54 miles (43,194) Joliet – 20 miles (148,262) Kankakee – 21 miles (26,445) Peoria – 81 miles (114,265) Chicago – 53 miles (2.7 mil)	2883
Byron Generating Station	Byron, IL	49.4 °F	Joliet – 73 miles (148,262) Quad Cities – 74 miles (474,937) Rockford – 17 miles (147,651) Milwaukee – 97 miles (595,047) Chicago – 86 miles (2.7 mil)	2917
Dresden Generating Station	Morris, IL	50.1 °F	Chicago – 43 miles (2.7 mil) Dekalb – 64 miles (43,194) Joliet – 25 miles (148,262) Kankakee – 34 miles (26,445)	2883
Calvert Cliffs	Lusby, MD	57.7 °F	Baltimore – 60 miles (621,849) Washington, D.C. – 46 miles (681,170) Dover, DE – 70 miles (37,786) Richmond, VA – 82 miles (220,289) Waldorf – 29 miles (67,752)	3343

In

Crop	Average lbs/ft^2	Average \$/ft^2	Average price per lb	Optimal temp. range	\$/ft^2/year	# of crop cycles/year (high end)
Tomato	1.81	\$5.95	\$3.29	60-80°F	\$11.90	2
Cucumber	0.37	\$0.48	\$1.30	60-85°F	\$1.44	2-3

Lettuce	0.47; Romaine - 0.71	$0.86-1.90	Romaine - $1.84-2.75; Spinach - $3.83	60-75°F	$11.40	6
Saffron	0.13 g	$4.03-5.32	$1500-9000; $20/g	63-77°F	$21.28	4
Basil	0.2 lbs/plant	$1.60-12.00	$8-60	45-80°F	$60.00	5
Medical Cannabis	Outdoor - 2.5 lb/plant; Indoor - 0.6 lb/plant; Mixed - 0.89 lb/plant	$1402-3505	Outdoor – $1402 Indoor – $2275 Mixed – $1575	60-95°F	$9,800	4
Button	6.59	$6.75-11.38	$1.31	50-77°F	$79.66	7
Geraniums	N/A	$12.00/flat	N/A	45-80°F	$12.00	1
Poinsettia	N/A	$18.00/3.5 quart pot	N/A	60-75°F	$18.00	1
Oyster	11-15	$110-115	$3.10; $6-12	36-77°F	$115	1

Table 5, information is compiled regarding average returns on select crops. This does not account for planting density, crop variety, USDA growing zone, or management practices. It should also be noted that year-round production of tomatoes is not always feasible due to heating and lighting requirements, harvest time of basil is based on anticipated use of the herb, and flower crop cycles are dependent on market and plant holder type (e.g. pot, tray, hanging basket) making them a difficult price point to estimate. It should also be noted that the unusual legal status of medical cannabis may raise more complicated economic concerns than represented here. Table 6 illustrates the potential returns from large greenhouses growing certain crops. The largest greenhouse in the United States is 21,000,000 ft^2 (Color Spot Nurseries, CA) and the largest greenhouse in Illinois is 2,013,000 ft^2 (N.G. Heimos Greenhouse).

Crop	Average lbs/ft^2	Average $/$ft^2$	Average price per lb	Optimal temp. range	$/$ft^2$/year	# of crop cycles/year (high end)
Tomato	1.81	$5.95	$3.29	60-80°F	$11.90	2
Cucumber	0.37	$0.48	$1.30	60-85°F	$1.44	2-3

Lettuce	0.47; Romaine - 0.71	$0.86-1.90	Romaine - $1.84-2.75; Spinach - $3.83	60-75°F	$11.40	6
Saffron	0.13 g	$4.03-5.32	$1500-9000; $20/g	63-77°F	$21.28	4
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Medical Cannabis	Outdoor - 2.5 lb/plant; Indoor - 0.6 lb/plant; Mixed - 0.89 lb/plant	$1402-3505	Outdoor – $1402 Indoor – $2275 Mixed – $1575	60-95°F	$9,800	4
Button	6.59	$6.75-11.38	$1.31	50-77°F	$79.66	7
Geraniums	N/A	$12.00/flat	N/A	45-80°F	$12.00	1
Poinsettia	N/A	$18.00/3.5 quart pot	N/A	60-75°F	$18.00	1
Oyster	11-15	$110-115	$3.10; $6-12	36-77°F	$115	1

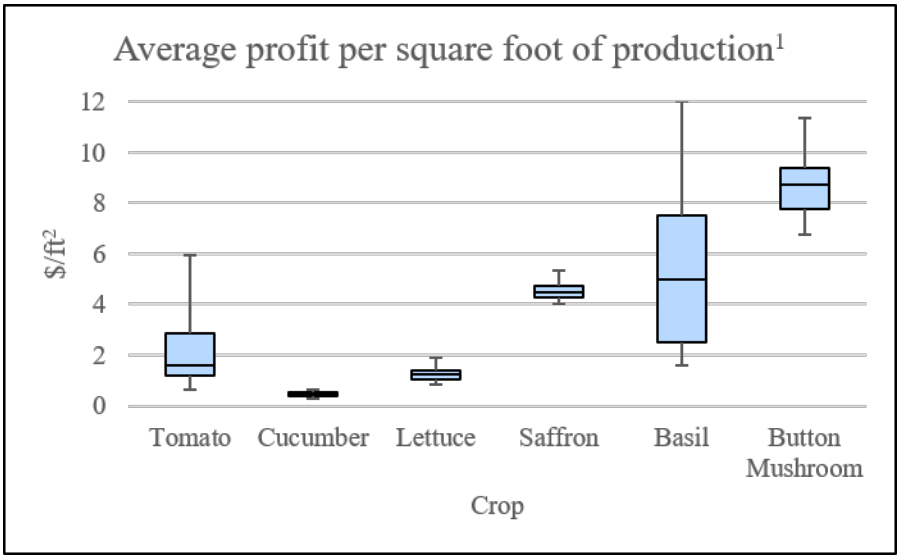
Table 5 Average returns on selected crops

Table 6 Projected returns from large greenhouses by crop type

Crop	**$/ft^2/year**	**M$/year**		
		Largest Greenhouse in IL [a]	**Largest Greenhouse in US [b]**	**1 square mile [c]**
Tomato	$11.90	$24	$250	$331.75
Cucumber	$1.44	$3	$30	$40.14
Lettuce	$11.40	$23	$239	$317.81
Saffron	$21.28	$43	$447	$593.25
Basil	$60.00	$121	$1,260	$1,672.70

Cannabis	$9,800	$19,727	$205,800	$273,208.32
Button	$79.66	$160	$1,673	$2,220.79
Geraniums	$12.00	$24	$252	$334.54
Poinsettia	$18.00	$36	$378	$501.81

[a] Largest Greenhouse in IL: 2,013,000 ft^2 (N.G. Heimos Greenhouse, IL)
[b] Largest Greenhouse in US: 21,000,000 ft^2 (Color Spot Nurseries, CA)
[c] 1 square mile ~ 27,878,400 ft^2 (potential effective zone)



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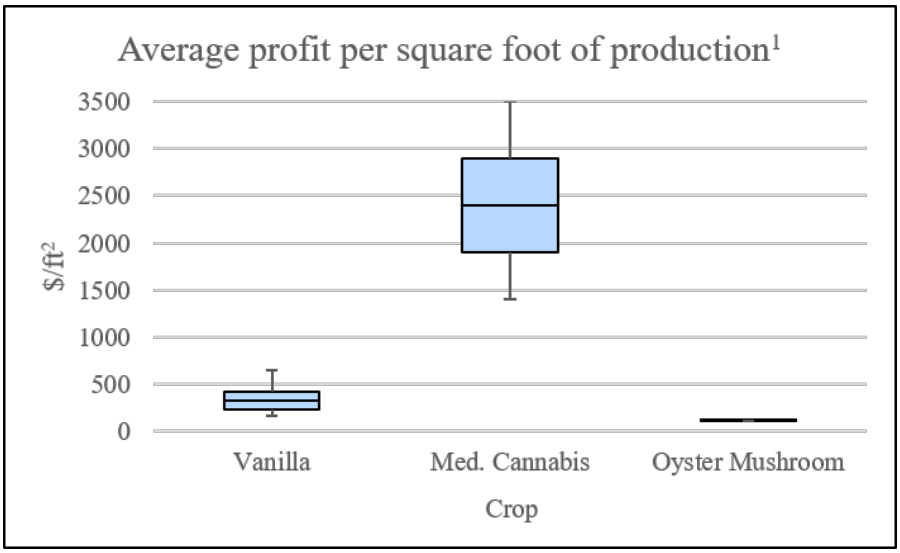

(b)

Figure 6 Average profit per square foot of production for (a) conventional and (b) non-conventional greenhouse crops.

[1] The average profit/ft^2 of production for all crop candidates except pineapple ($1.26/pineapple) and avocado ($246-802/tree). Profit ranges do not consider agronomic factors such as management practices, thus, prices may be higher under actual production conditions.

4.2. Economic considerations of waste heat requirements

The objective of the following analysis is to explore factors important to the economic viability of the waste heat uses from Exelon NPPs for agricultural applications. This evaluation focuses on the use of discharge water of NPPs to supply heat to greenhouses because this is identified as a technically and economically promising agricultural application. To determine the possible profits of selling NPP discharge water as a heat source, it was first considered what a grower's typical heating costs are in a commercial greenhouse. For the calculations, a greenhouse model unit was considered as shown in Figure 7. In the diagram, *L* represents the length of the greenhouse, *W* the width, *S* the height of the sidewall, and *H* the total height. These values were L=96ft, W=30ft, S=5ft, and H=8ft. The material of the walls was 4 mm (5/32") of twin-wall polycarbonate with a U-value of 0.7.

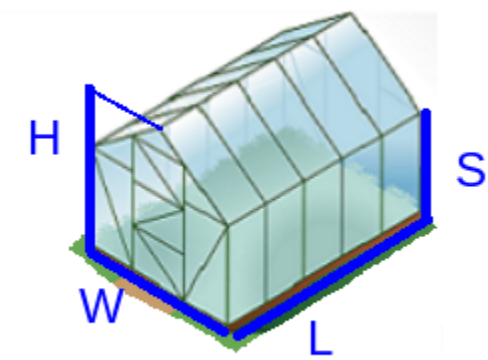

Figure 7 Greenhouse model unit (earthcaregreenhouse.com, 2015).

The heating cost was calculated under the assumptions that the greenhouse principal product was basil and that the target greenhouse temperature was constantly at 62 °F, which is the average temperature between the maximum and minimum temperature needed for this high-value crop, and that there is no considerable contribution from the amount of incoming solar radiation and from the heat loss linked to air leakages in the

structure. Hence, the heat required to maintain the target temperature of such a greenhouse for 1 hour was calculated using the following formula

$$E[Btu/h] = U\left[\frac{Btu}{hr\ ft^2\ °\mathrm{F}}\right]A_e[ft^2](T_{inside}[°\mathrm{F}] - T_{outside}[°\mathrm{F}]) \quad (1)$$

where A_e is the total exposed area (area of the greenhouse walls in contact with the outside air) of 4,287 ft².

In this case study, the greenhouse unit was located near the Calvert Cliff NPP in Maryland to capture similar climatic conditions as a possible NPP waste heat-based greenhouse. The average temperature of that region throughout the year is displayed in Figure 8. The data was measured in a meteorological station at the Baltimore Washington International Airport. The heat load for the greenhouse unit model was calculated and is shown in Figure 9.

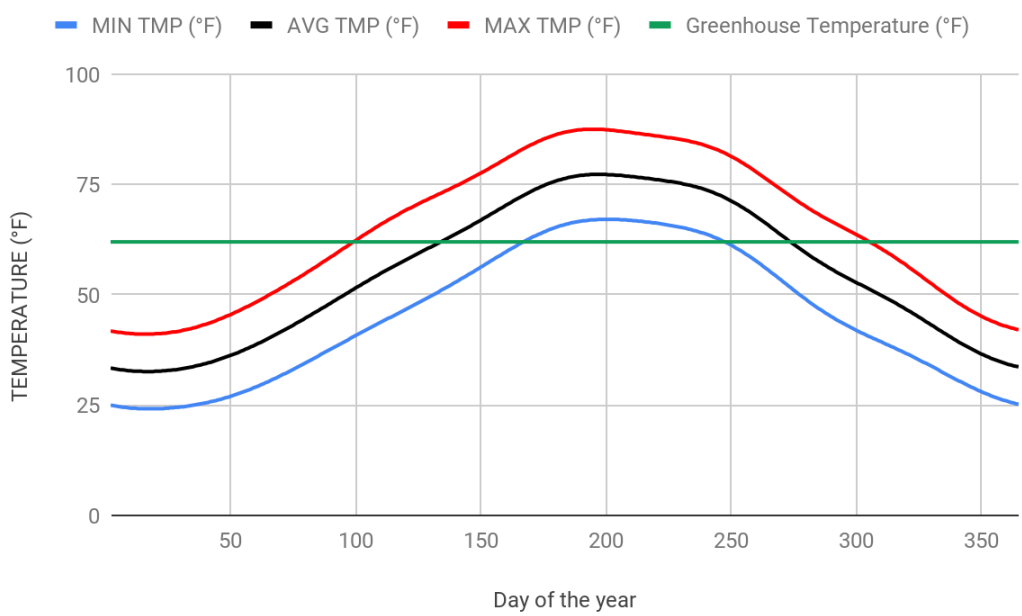

Figure 8 Average temperature at Baltimore Washington International Airport, MD (National Center for environmental Information, 2018).

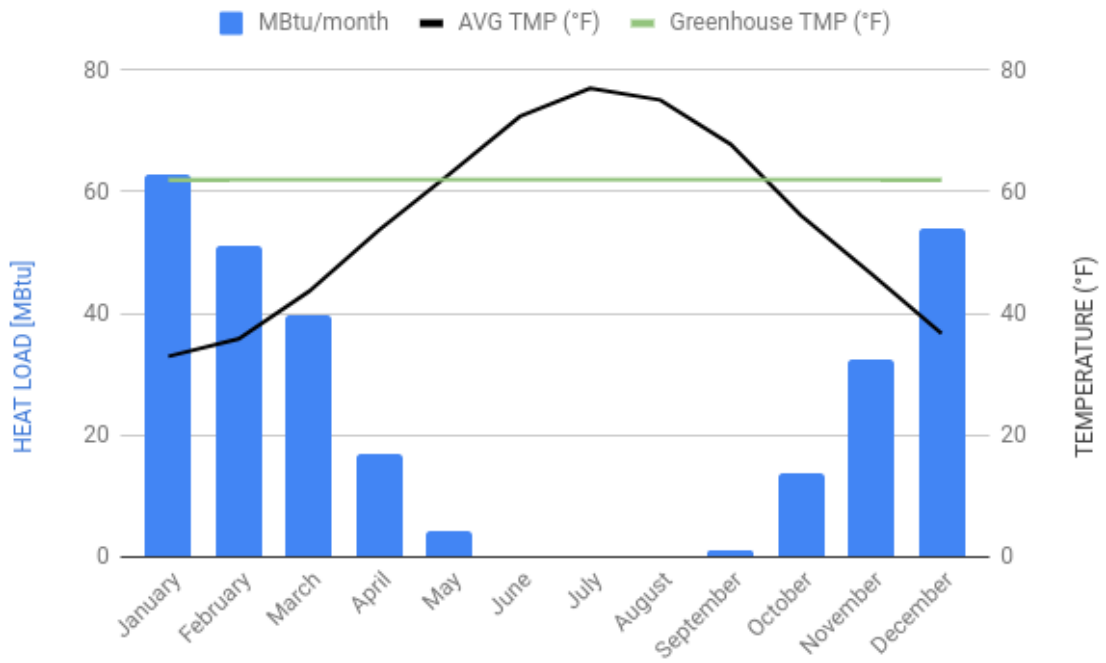

Figure 9 Heat load per greenhouse model unit (2880 ft^2) throughout the year.

To determine the cost to fulfill the heat requirements (heat load) it was necessary to choose a fuel and a heating appliance, which are analyzed in Table 7 Heating fuel comparison (EMC, 2013; U.S. Energy Information Administration, 2018). The prices per unit shown in the table are those corresponding to the prices for the South Atlantic coast of the US in 2017. As the data from the U.S. Energy Information Administration is given for different months, those values were averaged over one month to simplify the calculations.

From Table 7 different heating appliances were chosen for the different fuels. In each case, the boiler is chosen but in the case of the electricity the air-source heat pump is also considered. In Figure 10 the annual cost for heating the greenhouse unit model for such appliances and types of fuel is shown. In the case of electricity, Electricity 1 represents the air-source heat pump while Electricity 2 represents the boiler.

Table 7 Heating fuel comparison (EMC, 2013; U.S. Energy Information Administration, 2018).

Fuel Type [Fuel Unit]	Fuel Price per Unit (dollars)	Fuel Heat Content Per Unit (Btu)	Fuel Price per MBtu (dollars)	Heating Appliance Type	Approx. Efficiency (%)	Fuel Cost per MBtu ($)
Natural gas [Them]	0.9605	100,000	9.61	Furnace or Boiler	82	11.71
				Room Heater (Vented)	65	14.78
				Room Heater (Unvented)	100	9.61
Propane [Gallon]	3.2367	91,333	35.44	Furnace or Boiler	80	44.30
				Room Heater (Vented)	65	54.52
Electricity [KWh]	0.1013	3,412	29.69	Furnace or Boiler	98	30.30
				Air-Source Heat Pump	240	12.37
				Geothermal Heat Pump	330	9.00
				Baseboard/Room Heater	100	29.69
Fuel Oil #2 [Gallon]	2.6918	138,690	19.41	Furnace or Boiler	78	24.88

[1] Natural gas is typically sold to residential customers in units of "therms", but it may be sold in units of cubic ft. One therm is equivalent to about 97.847 cf.

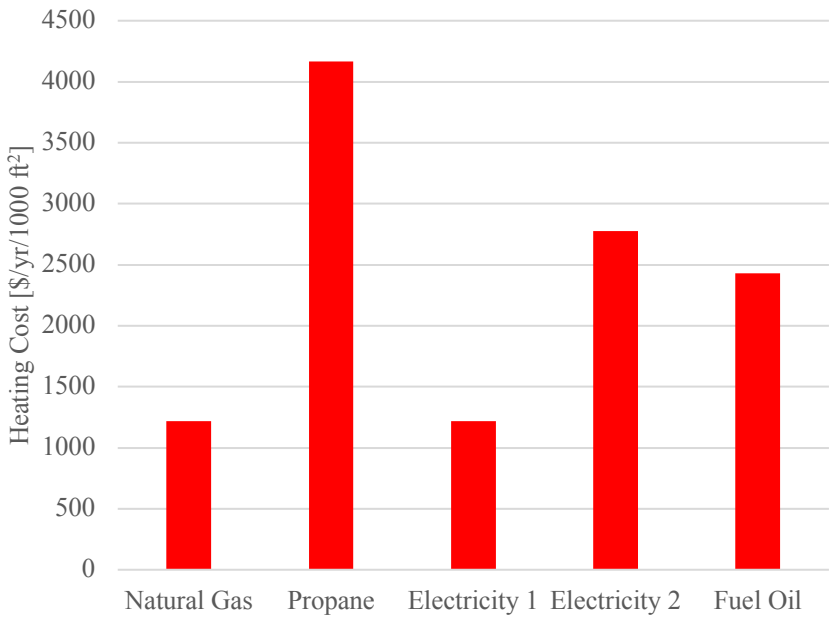

Figure 10 Annual cost for heating for different fuels per 1000 ft^2 of greenhouse area. Electricity 1 considers an air-source heat pump as the heat appliance while the rest of the cases consider boilers.

Natural gas is one of the most economical fuels, although it is not available to growers in all areas. It needs no on-site storage as it is piped from transmission lines. Adding the transmission cost through the pipelines will give the total fuel cost. Although Propane (LPG) is more expensive than natural gas it can be readily obtained where natural gas is not available. Maintenance is minimal, but a storage tank and preheater are needed (Bartok, 2005).

Fuel Oil No. 2 may be more expensive in some locations especially where it must be transported a long distance from the supply point. It is a fuel that demands slightly more burner maintenance than gas. Oil requires on-site, above ground storage tanks that must have a containment in event of a leak or spill. Oil stored in outdoor, above-ground tanks may become difficult to pump in temperatures near 0 ºF (Bartok, 2005).

A heating system using an alternative fuel can be installed to meet the total heat needs of the greenhouse but it is generally too expensive. Systems that burn fuels, such as

wood, coal and waste oil require more attention and labor. Maintenance costs may also be greater and should be readily available (Bartok, 2005).

Applying this fuel cost information shows how expensive heating can be. Take heated greenhouses of the size found at Bugey for example. Bugey covered 12.3 acres, which is over 520,000 ft^2 (CEC, 1988). To heat such a greenhouse with propane would cost in excess of $2.1 million per year according to this analysis. Even cheaper heating options, like natural gas or the electric air-source heat pump would cost about $650,000 per year. It should be emphasized that this is a very rough estimate and a more detailed analysis which includes specific greenhouse architecture is required to provide a realistic estimate on greenhouse heating costs. This analysis also considers that the greenhouse is kept at the mean growing temperature without consideration of minimum growing temperature or improved growth potential at elevated temperatures.

4.2.1. Analysis for different crops

As seen in the previous section, the costs will depend on several factors, such as the temperature inside the greenhouse, which in turn depends on the crop, and the temperature outside. For this reason, the heating cost per square foot for a year for a greenhouse unit in Illinois and in Maryland is calculated using the greenhouses located near Rockfield, IL, and Calvert Cliff, MD, respectively. Electricity was chosen as the heat source with an air-source heat pump as the heating appliance. The results are displayed in Figure 11.

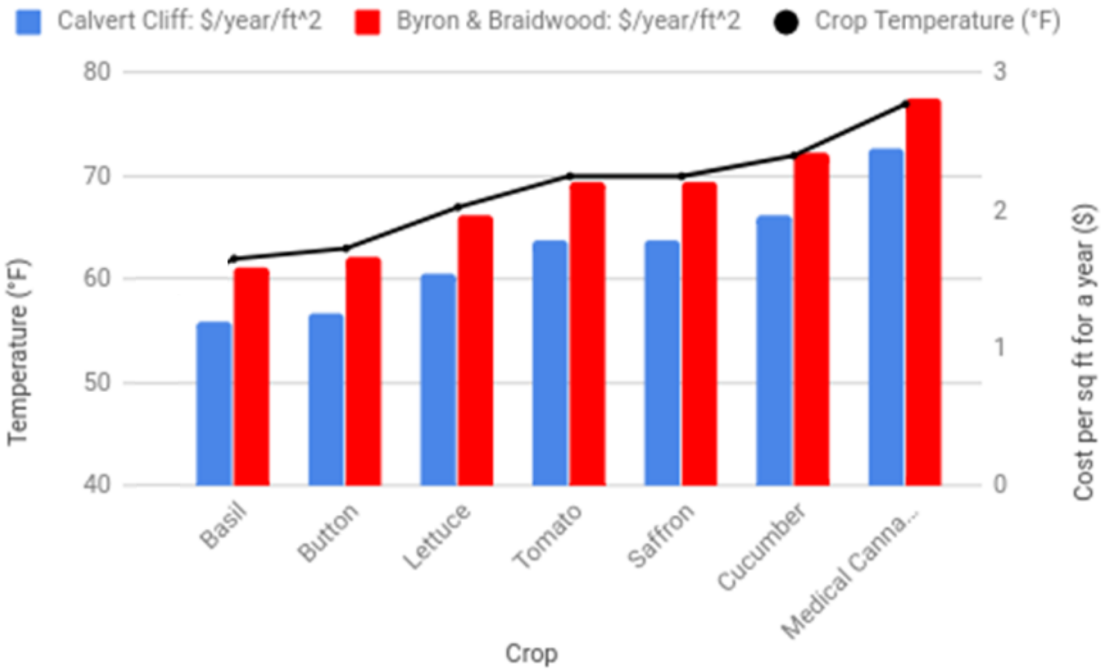

Figure 11 Cost per square foot for a year for different crops and locations of the units.

4.2.2. Horticulture Scale Analysis

The following section estimates the potential size of the horticultural area when utilizing the waste heat of the Calvert Cliff NPP. Such area was calculated based on the peak heating load of the greenhouse model unit, which was the amount required to heat the greenhouse during the coldest hour of the coldest night of January. That amount results in Q_L= 30.38 kW and corresponds to an outside air temperature of 27.5 °F. The available heat (Q_s) was calculated as the product of the mass flow rate and sensible heat. Finally, dividing Q_s by Q_L it was possible to determine the number of greenhouse model units that can be fed. Then, multiplying that number by the footprint area of the greenhouse model, which was A=2880 ft^2, the area that can be supplied was determined.

The condenser outlet temperature, displayed in Figure 12, was calculated knowing the temperature rise of the condenser, shown in Table 3, and the temperature of the Chesapeake Bay. As the temperature of the bay changes throughout the year, the difference of temperature between the outlet of the condenser and the greenhouse will change as well,

varying the number of greenhouse units that could be heated. The lower limit of heating capability is in principle when the waste heat supply temperature is equal to the desired greenhouse temperature.

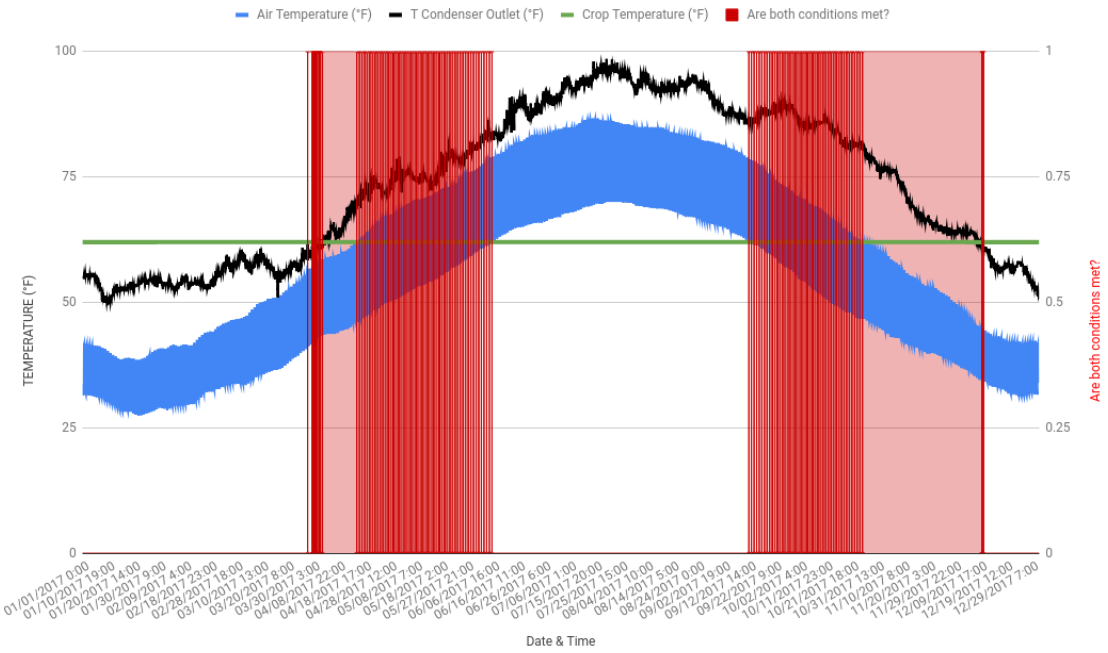

Figure 12 Air temperature, condenser outlet temperature, and hours when conditions to use NPP waste heat to warm up greenhouse are met. (National Ocean Service, 2018)

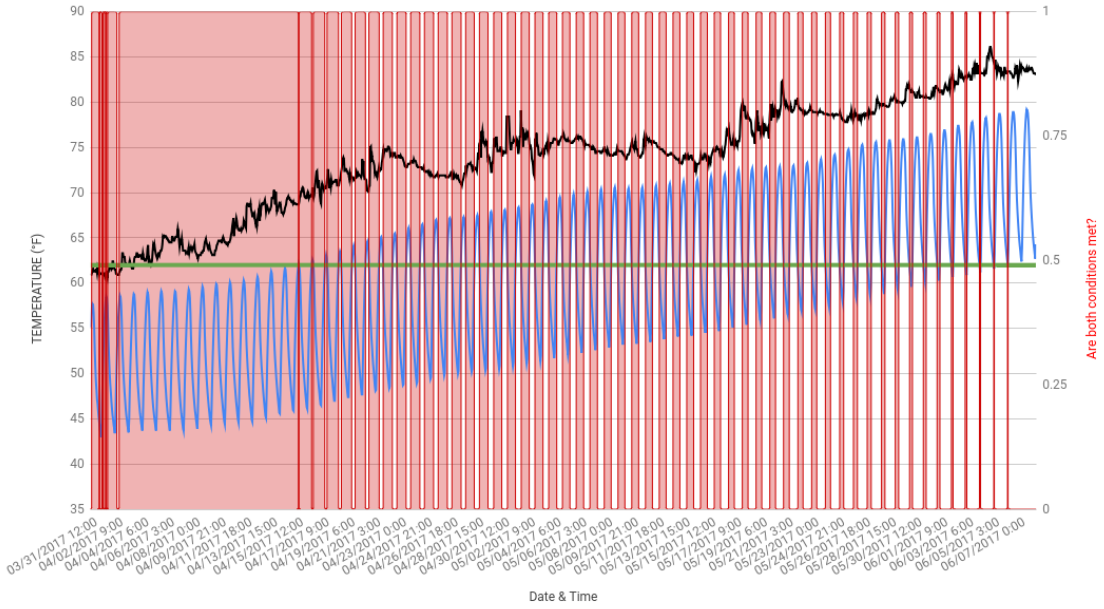

Figure 13 Zoom into Figure 12 graph: March 31st, 2017 – June 7th, 2017

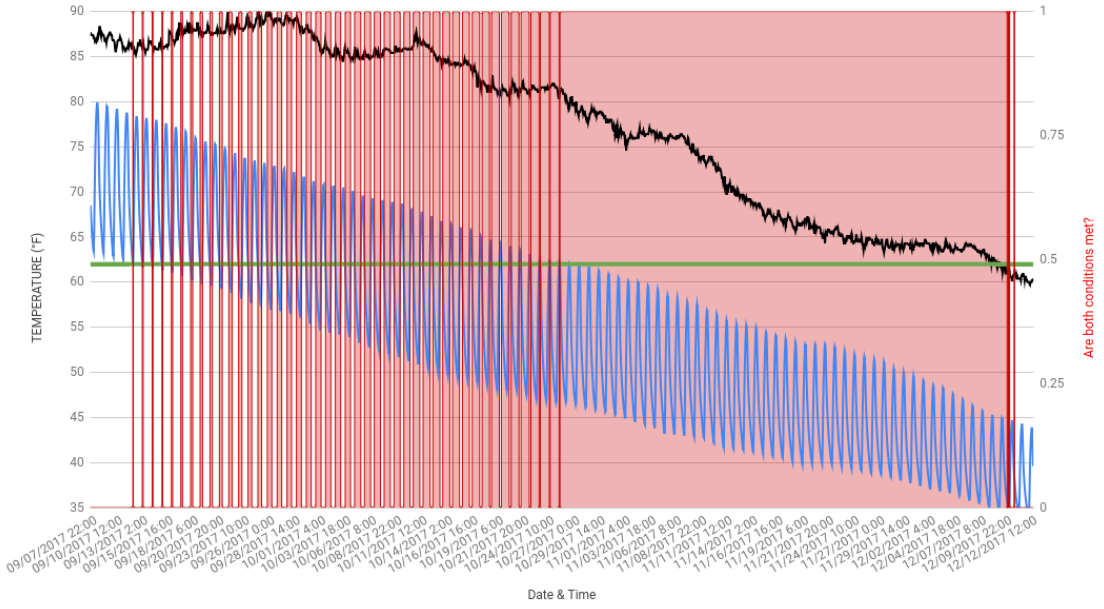

Figure 14 Zoom into Figure 12 graph: September 7th, 2017 – December 12th, 2017

Two conditions to determine the usability of waste heat are considered. First, that the outlet temperature of the condenser was 1 degree higher than the greenhouse temperature.

Second, that the unit needed the heat, that is, the air temperature was lower than the greenhouse temperature. In Figure 12, the hours when these conditions were met are shown. When the value of the curve is 1 it means that the conditions are met, while 0 the opposite. Both conditions were met for 2613 hours during a year. It should be noted that those hours were not consecutive. In other words, both conditions are met for some hours a day (usually in the night), and on the next days for some other hours. On Figure 13 and Figure 14, more detailed graphs of Figure 12 are shown. The minimum total amount of greenhouse units that could be heated by Calvert Cliff NPP was 5732, leading to an area of 16.5 million ft^2.

An assumption in these calculations was that there were no heat loses in the greenhouse model unit due to air leakages. A second one was that the incoming radiation does not have a considerable impact over the inside temperature. Thirdly it was assumed that the greenhouses located in the calculated area were not in contact to each other which would decrease the exposed area of the greenhouses, reducing the required heat load. Such assumptions can be redefined in further studies as their contributions have significant impact over the heat requirements of the greenhouse, but depend on the particular conditions and architecture of the greenhouses considered.

Waste heat from the Calvert Cliff NPP may be a dependable source of heat for the greenhouses depending on waste heat coupling to greenhouse production systems and the architecture of these greenhouses. However, another application could use the discharge water as preheated water, then heat it up in the greenhouse to achieve the desirable temperature and meet the heat requirements. In such way, the cost for running a greenhouse will decrease considerably. Evaluation technical and economic feasibility of the Calvert Cliff NPP in a future detailed assessment is necessary to fully determine potential waste heat re-use at this more complex site.

It is important to note that it is not fully clear where the temperature rise of the condenser at the Calvert Cliff NPP was measured. It appears to be the difference between the bay temperature and the discharge water temperature as the temperature rise of 12 °F coincides with the maximum temperature rise allowed by the EPA regulations. As a consequence, the actual outlet temperature of the condenser should be higher as the water

coming out of the condenser is mixed in a surge pit with bay water before coming out to the bay again. This characteristic will make this study and underestimated case of the real conditions in the plant.

In the case of the selected Illinois NPPs, future research should target the coupling of waste heat sources with greenhouses and the reciprocal consideration of greenhouse architectures. The Byron and Braidwood NPPs have cooling towers, resulting in higher condenser outlet temperatures which may allow them to provide waste heat at a temperature more easily used in agricultural applications than the Calvert Cliffs NPP. This reinforces the idea of having more accurate information of the condenser outlet temperature for developing a more detailed site-specific evaluation of agricultural possibilities at Calvert Cliffs NPP. Heat loss through the pipeline is another site-specific consideration requiring detailed analyses that considers specific greenhouse architectures and high-value crop choices. There will be heat transfer occurring between the water and the soil, decreasing the usable heat. Further studies can take this phenomenon into account to determine how much heat is lost through the pipes for given distances.

CHAPTER 5. REGULATORY CONSIDERATIONS FOR WASTE HEAT USES

There are few regulations that clearly affect use of discharge water from a NPP. The waste heat from a nuclear reactor has not been used in the United States, so the NRC has had no reason to implement regulations, though it reserves the right to do so. This lack of regulation should not be interpreted as a free pass, however, as other government agencies such as the FDA and EPA may intervene if a waste heat project is implemented. The NRC, FDA, and EPA all may share some jurisdiction of waste heat utilization based on the target application.

5.1. Overview of regulatory factors

The NRC is the main regulator of nuclear power in the United States. Any changes to the nuclear generating system are subject to the NRC's approval which may be relevant for diverting discharge water. Waste heat would be taken as the discharge from the condenser, which is usually treated to be suitable for release as groundwater. Groundwater is the usual water source for a traditionally heated greenhouse as well as the destination for waste heat water, so any regulations regarding groundwater release should be assumed applicable to a waste heat project. Information is already made public through the NRC regarding the content of waste from NPPs, making it simple to demonstrate whether use of waste water is safe.

Table 8 Liquid effluent radiation release

	Braidwood	**Byron**	**Dresden**	**Calvert Cliffs**
Continuous Liquid Effluent Tritium Release [Ci/a]	116	88.2	Only listed total release	0.704
Batch Liquid Effluent Tritium Release [Ci/a]	2310	900	Only listed total release	1279
Total Liquid Effluent Radioactive Release [Ci/a]	2426	988	0.176	1280
Avg Diluted Conc. [10^{-5} µCi/mL]	4.47	6.99	0.186×10^{-6}	2.75×10^{-2}
Percent EPA Limit (Drinking Water)	22.4%	35.0%	~0%	0.138%
Percent NRC Limit (Unrestricted Use)	4.47×10^{-3} %	6.99×10^{-3} %	~0%	2.75×10^{-7} %
Braidwood Nuclear Power Station Annual Radioactive Effluent Release Report (ARERR) (2017)				

Tritium is by far the most prevalent radioactive substance in the liquid effluent from the plant, with other radioisotopes at concentrations often undetectable. By EPA standards, 200 pCi/mL is the allowable limit for tritium in drinking water, so the effluent from NPPs is considered safe to drink from a radiological point of view. The NRC regulated limit of 1 Ci/L leaves an even more significant margin for use with each plant producing thousandths of a percent of that value. It is noticeable that despite the low average diluted concentrations, most radiation is released in batches. It may be desirable to implement barriers between batch releases and waste heat uses (NRC, 2017a; NRC, 2017b; NRC, 2017c; NRC, 2017d).

The EPA, like the NRC, will primarily be interested in establishing that there are no unsafe releases of radioisotopes to the surroundings. This is not expected to be an issue as the water treatment and releases should be the same as they were prior to waste heat utilization. The only difference will be in the routing of the waste heat water through agricultural systems which may increase the risk of a break in the water line causing unnecessary exposure to radioisotopes, like tritium.

The FDA becomes relevant as soon as waste heat is used to produce food or medicine. Interestingly, there are few regulations regarding exposure of food to radioactive material. Radiation is often used to kill potential parasites in food at a much higher dose than waste water could achieve. This radiation exposure is considered safe and does not have to be listed on packaging (Section 402(a)(7) of the FD&C Act), but intentional radiation sources in other cases must be listed as an additive ingredient (Section 201(s) of the FD&C Act). Taking these rules in consideration within the scope of a waste heat project, the FDA's approval of waste heat utilization as meeting safe conditions of use may be desirable. Any exposure to radiation from waste heat water should be negligible, especially without direct contact, so any barrier from the FDA should not pose much challenge.

5.2. Considerations for potential agricultural applications

Due to the lack of existing regulations, it is difficult to determine what considerations need to be made from the regulatory perspective. It is established that the overall aim of regulations is to protect the safety, health, and welfare of the public and environment, so any applications of waste heat naturally must take this into consideration. It would make sense to take a conservative approach to ensure that there are many barriers between agricultural processes and power plant activity. Fortunately, existing regulations and designs of NPPs offer several barriers to any dangerous exposure, leading to releases of radioactive material well below thresholds considered to verge on unsafe.

Some NPPs have come under controversy for unintentional leaks of radioactive material, so safeguards against these events may be the only need for agricultural applications. While it would take an extremely serious leak to make crops potentially unsafe for production, it is important to follow the spirit of regulation and separate the public and environment to the greatest extent reasonably possible. For this reason, the agricultural system should be able to run on heat from backup generators in an emergency, and it should be possible to isolate the agricultural system from the power plant waste heat.

CHAPTER 6. RECOMMENDATIONS AND CONCLUDING REMARKS

The waste heat from Exelon's fleet of NPPs is a plentiful yet untapped source of energy. Whether a power plant is dedicated to electricity generation or is already providing heat through a combined heat and power scheme, waste heat utilization should clearly be considered for its potential to significantly improve agricultural profitability.

While waste heat is produced at every nuclear power plant, this does not necessarily mean every NPP is ready or able to implement waste heat utilization in agricultural applications. The NPP should be in an area with a potential demand for waste heat, provide water at a temperature that will have an appreciable impact on heating costs, meet all potential regulatory requirements regarding water quality, and expect to operate far enough into the future to be considered a reliable long-term asset to those using the waste heat. All these considerations except for the first may be corrected at will and are rarely an issue for NPPs. For this reason, it should be re-emphasized that the existence of a waste-heat-utilizing system will not necessarily draw consumers of waste heat, so the application of waste heat must be selected to meet a demand.

Fortunately, high-value agriculture production has been demonstrated to benefit from waste heat. There are other means of using the waste heat, such as heat irrigation and open-field soil heating, but climate-controlled greenhouse standout as a proven beneficiary of waste heat and with higher potential both technically (e.g., crop type) and economically. Heating costs account for a large share of the expenses of a climate-controlled greenhouse, so the elimination or even reduction of this expense will make them a much more attractive option for growers in the cold climates in which Exelon NPPs are situated. Interested growers may be drawn by the reduced heating costs, the improved crop yields associated with climate-controlled greenhouses, and the potential high-value and/or novel crops that may be grown under conditions of plentiful and low-cost heating. Illinois already depends

on nuclear power for a large portion of its electricity, tax revenue, and employment. More industries, tax revenue, and employment will depend on Exelon's NPPs than before if waste heat is utilized successfully as a commodity in regional, high-value agricultural industrires.

To assess economic possibilities of these technically feasible options, greenhouse-scale pilot evaluations for waste heat utilization for specific high-value crops is vital. This will require analysis of the demands on the nuclear and agricultural side of such an undertaking. Thus far it has been shown that waste heat utilization in agriculture is viable and beneficial to all parties involved, but this has been achieved by several different methods. A waste heat system developed with an Exelon plant and specific crops in mind will be necessary to gauge if such a project is viable at this time.

The next phase of this project will require modeling of a coupled nuclear power generation and heated greenhouse system. Such a model will allow for potential sites to be chosen and for the needs of a greenhouse to be determined. This will be instrumental to determining such matters as what crops will be viable, how much waste heat could be sold, how much back-up heating capacity will be required, and what greenhouse heating strategies will work best. Potential crops will require a more detailed analysis with special attention paid to non-traditional, high-value crops such as spices and medical cannabis. At this point, with more detailed information regarding waste heat needs and temperatures as well as crops chosen, it will be possible to provide a detailed economic analysis of what returns can be expected from this system. While greenhouses operating with aid of waste heat have been commercially successful in the past, they have also failed; in the case of Bugey the same greenhouse system failed and was successful under different management. Knowing this, a detailed economic analysis in addition to a clear model of the system will be necessary for the success of a waste heat reutilization project.

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